

LDL-cholesterol concentrations: a genome-wide association study

SCIFACT-EVIDENCE-2095573

BACKGROUND LDL cholesterol has a causal role in the development of cardiovascular disease. Improved understanding of the biological mechanisms that underlie the metabolism and regulation of LDL cholesterol might help to identify novel therapeutic targets. We therefore did a genome-wide association study of LDL-cholesterol concentrations. **METHODS** We used genome-wide association data from up to 11,685 participants with measures of circulating LDL-cholesterol concentrations across five studies, including data for 293 461 autosomal single nucleotide polymorphisms (SNPs) with a minor allele frequency of 5% or more that passed our quality control criteria. We also used data from a second genome-wide array in up to 4337 participants from three of these five studies, with data for 290,140 SNPs. We did replication studies in two independent populations consisting of up to 4979 participants. Statistical approaches, including meta-analysis and linkage disequilibrium plots, were used to refine association signals; we analysed pooled data from all seven populations to determine the effect of each SNP on variations in circulating LDL-cholesterol concentrations. **FINDINGS** In our initial scan, we found two SNPs (rs599839 [$p=1.7 \times 10^{-15}$] and rs4970834 [$p=3.0 \times 10^{-11}$]) that showed genome-wide statistical association with LDL cholesterol at chromosomal locus 1p13.3. The second genome screen found a third statistically associated SNP at the same locus (rs646776 [$p=4.3 \times 10^{-9}$]). Meta-analysis of data from all studies showed an association of SNPs rs599839 (combined $p=1.2 \times 10^{-33}$) and rs646776 ($p=4.8 \times 10^{-20}$) with LDL-cholesterol concentrations. SNPs rs599839 and rs646776 both explained around 1% of the variation in circulating LDL-cholesterol concentrations and were associated with about 15% of an SD change in LDL cholesterol per allele, assuming an SD of 1 mmol/L. **INTERPRETATION** We found evidence for a novel locus for LDL cholesterol on chromosome 1p13.3. These results potentially provide insight into the biological mechanisms that underlie the regulation of LDL cholesterol and might help in the discovery of novel therapeutic targets for cardiovascular disease.